LFM Pulse Compression, Link Budget Modeling & +10 dB SNR Enhancement for Long-Range Radar Target Detection

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I. PROBLEM STATEMENT & MATHEMATICAL FORMULATION

A. Radar Energy vs. Resolution Conflict: In long-range orbital tracking and launch vehicle surveillance at SDSC SHAR (target range R = 200 km, RCS $\sigma = 1.0\ m^2$), detecting weak echoes requires maximizing radiated pulse energy $E_t = P_t \cdot \tau_p$. For unmodulated rectangular pulses, range resolution is constrained by pulse duration: $\Delta R_{unmod} = c \cdot \tau_p / 2$. Widening an unmodulated pulse from 10 µs to 100 µs scales energy by +10 dB but degrades range resolution from 1.5 km to 15.0 km, rendering target discrimination impossible. Applying Linear Frequency Modulation (LFM) decouples resolution from pulse width:

$$\Delta R_{\text{unmod}} = \frac{c\tau_p}{2} \quad \longrightarrow \quad \Delta R = \frac{c}{2B} = \frac{3.0 \times 10^8 \text{ m/s}}{2 \times 4.0 \times 10^6 \text{ Hz}} = \mathbf{37.50} \text{ m} \qquad (1)$$

B. LFM Chirp & Matched Filter Pulse Compression: A linear chirp sweeps frequency over sweep bandwidth B = 4.0 MHz within duration τ_p with chirp slope $k_r = B / \tau_p$. The receiver matched filter h(t) = s*(-t) compresses the echo into a sinc envelope whose Rayleigh resolution depends strictly on bandwidth B:

$$s(t) = \text{rect}\left(\frac{t}{\tau_p}\right) e^{j\pi k_r t^2}, \; k_r = \frac{B}{\tau_p} \quad \Longrightarrow \quad y(t) = s(t) * s^*(-t) \approx \tau_p \cdot \text{sinc}(\pi B t) \qquad (2)$$

C. S-Band Radar Link Budget: Target echo power P_r at R = 200 km (P_t = 250 kW, G = 40 dBi, λ = 0.1 m [3.0 GHz], L = 3.0 dB, F = 2.5 dB) via the Radar Range Equation yields:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R_0^4 L} = \frac{(250 \text{ kW})(10^4)^2 (0.1 \text{ m})^2 (1.0 \text{ m}^2)}{(4\pi)^3 (200 \text{ km})^4 \cdot 10^{0.3}} = 3.946 \times 10^{-14} \text{ W } (-134.04 \text{ dBW}) \qquad (3)$$

Receiver noise spectral density is $N_0 = k_B T_0 F$ = -201.48 dBW/Hz (7.120×10^{-21} W/Hz). Matched filter peak output SNR integrates total received energy $E_r = P_r \cdot \tau_p$. Comparing τ_2 = 100 µs against τ_1 = 10 µs proves the fundamental +10 dB SNR scaling law:

$$\text{SNR}_{\text{MF}} = \frac{P_r \tau_p}{N_0} \quad \Longrightarrow \quad \Delta\text{SNR}_{\text{MF}} = 10\log_{10}\left(\frac{\tau_2}{\tau_1}\right) = 10\log_{10}\left(\frac{100\ \mu s}{10\ \mu s}\right) = \mathbf{+10.00} \text{ dB} \qquad (4)$$

II. PROPOSED METHODOLOGY & RADAR PROCESSING PIPELINE

The simulation executes a discrete-time processing chain: (1) Dual LFM chirp generation (B = 4 MHz, τ_1 = 10 µs, τ_2 = 100 µs) at f_s = 16 MHz; (2) Pulse train synthesis at PRT = 2.0 ms (PRF = 500 Hz, R_{unamb} = 300 km); (3) Target propagation channel at R = 200 km (round-trip delay τ_d = 1.333 ms) with complex circular AWGN; (4) Zero-padded FFT cross-correlation matched filtering preventing circular wraparound; (5) 1-D CA-CFAR detection (48 reference cells, 16 guard cells, $P_{fa} = 10^{-6}$).

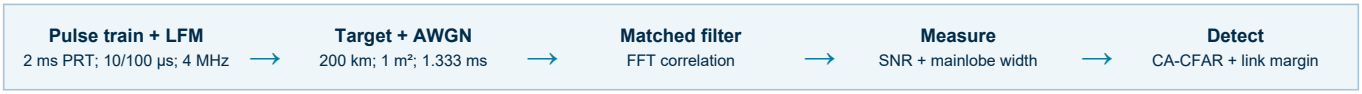


Fig. 1: End-to-end Track 1 simulation pipeline.

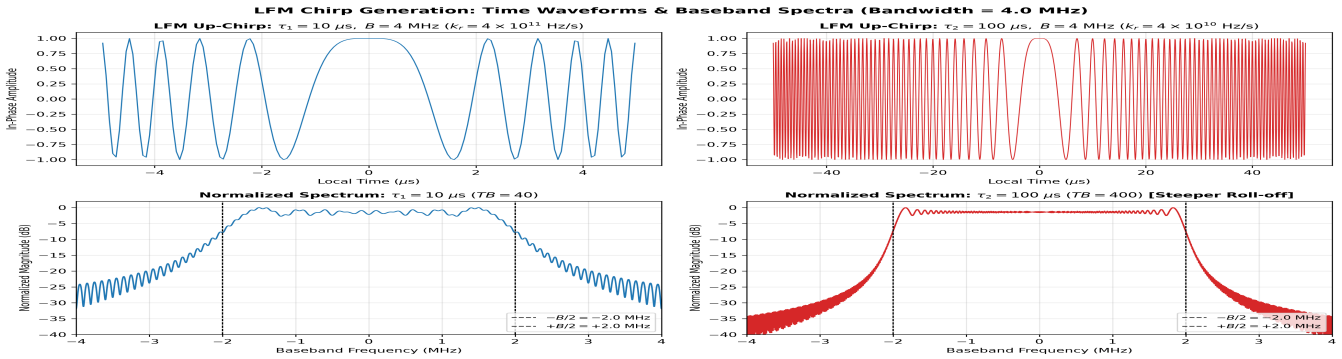


Fig. 2: Synthesized LFM chirp time-domain waveforms and normalized frequency spectra (B = 4.0 MHz) for 10 µs and 100 µs pulses.

III. QUANTITATIVE RESULTS, BENCHMARK COMPARISON & VALIDATION

Radar System Metric / Parameter	Pulse 1 (τ_1 = 10 µs)	Pulse 2 (τ_2 = 100 µs)	Enhancement / Comparison
Pulse Duration (τ_p)	10.0 µs	100.0 µs	×10.0 (+10.0 dB)
Duty Cycle (τ_p / PRT, PRT = 2 ms)	0.50 %	5.00 %	×10.0
Radiated Pulse Energy (P_t = 250 kW)	2.50 J	25.00 J	+10.0 dB
Chirp Sweep Slope ($k_r = B / \tau_p$)	4.00×10^{11} Hz/s	4.00×10^{10} Hz/s	1/10 slope
Time-Bandwidth Product ($TB = B \cdot \tau_p$)	40.0 (16.02 dB)	400.0 (26.02 dB)	+10.0 dB compression gain
Theoretical Matched-Filter SNR ($P_r \tau_p / N_0$)	17.44 dB	27.44 dB	+10.00 dB exact boost
Simulated Measured Peak SNR	17.13 dB	27.31 dB	+10.18 dB (Validates Theory)
Nominal Range Resolution c / (2B)	37.50 m	37.50 m	Identical Resolution (1.00×)
Measured Half-Power (-3 dB) Width	35.36 m	34.17 m	Similar widths
Peak Sidelobe Level (PSL)	-10.49 dB	-11.89 dB	Noisy-profile sidelobe statistic
Required SNR (P_D = 0.90, $P_{fa} = 10^{-6}$)	13.18 dB	13.18 dB	Marcum Q nonfluctuating model
Theoretical Link Margin	+4.25 dB (Marginal)	+14.25 dB (Robust)	+10.00 dB Link Margin Boost
CA-CFAR Target Detection Margin	+4.14 dB	+10.08 dB	+5.93 dB above adaptive thresh

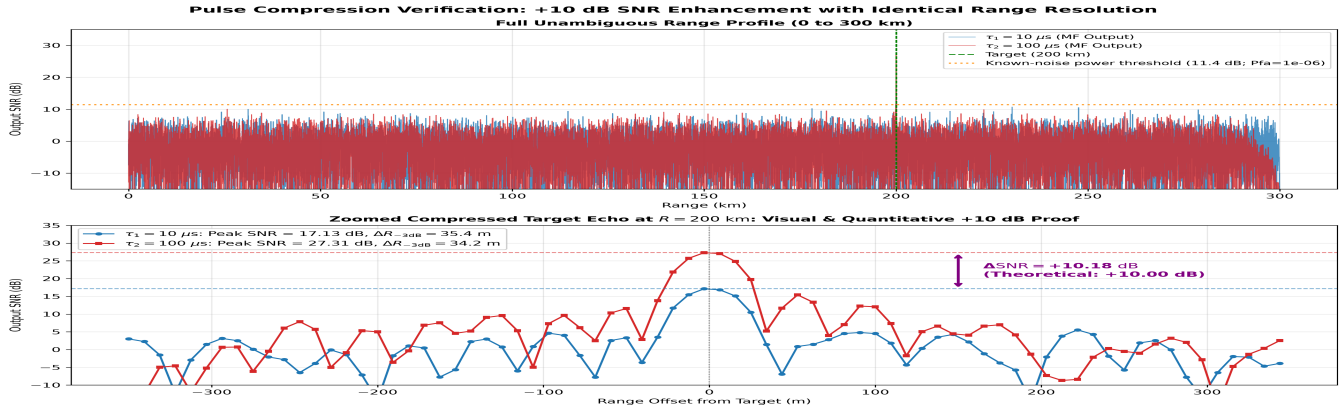


Fig. 3: Matched filter output at R = 200 km demonstrating the +10.18 dB SNR boost and similar 35 m mainlobe widths.

IV. KEY TAKEAWAYS, PRACTICAL CONSIDERATIONS & LIMITATIONS

1. Operational Significance: While a 10 µs pulse yields a marginal link margin (+4.25 dB) vulnerable to atmospheric attenuation, the 100 µs pulse provides a +14.25 dB link margin and +10.08 dB CFAR margin under the assumed link budget.
2. Minimum Range Blind Zone: Monostatic transmission blinds reception over $R_{min} = c\,\tau_p / 2$. For τ = 100 µs, R_{min} = 15.0 km (versus 1.5 km for 10 µs). Operational tracking radars employ dual-pulse schedules: short pulses for near-field acquisition and long chirps for orbital tracking.
3. Doppler Shift Coupling: Target radial velocity v_r induces Doppler frequency $f_d = 2v_r / \lambda$, causing an apparent range bias $\Delta R = -c\, f_d / (2\, k_r)$. Transmitting alternating up- and down-chirps cancels this bias in the ideal static-range chirp-pair model.
Validation: 200 production-chain trials: mean gain 9.99 dB; 95% CI for mean [9.88, 10.10] dB. Link parameters are assumptions, not verified radar specifications. CA-CFAR nominal Pfa assumes independent cells; this oversampled output is correlated. Peak SNR and sidelobes above are noisy single-run statistics.